

Evaluating Neoadjuvant Regimens for Rectal Cancer: A Single-Center Retrospective Study

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Introduction

The management of rectal cancer has undergone substantial changes in recent years, with neoadjuvant therapy emerging as a cornerstone in improving surgical outcomes and reducing recurrence rates. While standard neoadjuvant regimens typically involve chemoradiotherapy followed by surgical resection, there is growing variability in clinical practice as oncologists refine treatment strategies to improve outcomes. At LRH's Hollis Cancer Center, medical oncologists have employed a range of neoadjuvant regimens — some closely aligned with evolving standard-of-care protocols and others representing more novel approaches. This variability presents a unique opportunity to assess the comparative effectiveness of these regimens in achieving a complete pathological response (pCR), reducing recurrence rates, and potentially allowing patients to avoid surgery through a non-operative watch-and-wait strategy.

Aims

This study aims to compare the outcomes of different neoadjuvant regimens in patients with rectal cancer treated at Hollis Cancer Center from 2010 to 2025. By analyzing pCR rates, recurrence rates, and surgical outcomes, the study will evaluate which regimens are most strongly associated with improved clinical outcomes. Additionally, the study will explore the effectiveness of total neoadjuvant therapy in helping patients avoid surgery, a growing trend in rectal cancer management. Recognizing that treatment response may vary in patients with microsatellite instability-high (MSI-H) or mismatch repair-deficient (dMMR) tumors, we will also analyze outcomes in these patient subgroups to better inform individualized care strategies.

References



Design

The study will employ a retrospective cohort design, examining data from adult patients diagnosed with rectal cancer who received neoadjuvant therapy. Data collection will include demographic information, tumor staging, treatment details, surgical outcomes, pCR rates, recurrence data, and MSI/dMMR status. Patients will be grouped by neoadjuvant regimen to allow for meaningful comparisons. Statistical analyses will include chi-square and t-tests for categorical and continuous variables, respectively, as well as Kaplan-Meier survival analysis to evaluate recurrence-free survival. Multivariate logistic regression may be used to identify key predictors of pCR, recurrence risk, and surgical avoidance. This comprehensive design will ensure that meaningful conclusions can be drawn regarding regimen efficacy and patient outcomes.

Clinical Implications

This study holds significant potential to influence clinical practice by clarifying which neoadjuvant regimens provide the best outcomes for patients with rectal cancer. Identifying the most effective strategies for achieving pCR, reducing recurrence, and facilitating surgical avoidance could provide oncologists with actionable insights that improve patient care. By incorporating MSI-H and dMMR subgroup analysis, this research may also help refine personalized treatment strategies. Furthermore, by examining outcomes in patients treated with total neoadjuvant therapy, the study may contribute to a growing body of evidence supporting non-operative management in select patients.

Anticipated Challenges

Data collection will require manual retrieval. If the manual retrieval process becomes overwhelming, we are prepared to adjust the study timeframe to reduce the number of patients reviewed. For IRB approval, we are willing to modify our data requests and strongly believe our study will remain impactful. We will also collaborate closely with the surgical and medical oncology teams to ensure recurrence data is accurately interpreted.